1 $\frac{5-(-3)}{6-2}$ or equivalent **[M1]**

2 **[A1]**

[Total Marks: 2]

2 (a) Evidence of using $\frac{\text{change in } y}{\text{change in } x}$ with two correct points from the graph, e.g., $\frac{4-(-2)}{3-0}$ **[M1]**

2 **[A1]**

(b) $(0, -2)$ **[B1]**

[Total Marks: 3]

3 (a) Uses gradient formula: $\frac{-4-8}{4-(-2)}$ or $\frac{8-(-4)}{-2-4}$ **[M1]**

-2 **[A1]**

(b) Identifies coordinates of two clear points on the line, e.g., $(0, 1)$ and $(4, 4)$ **[M1]**

$\frac{3}{4}$ or 0.75 **[A1]**

[Total Marks: 4]

4 (a) Uses the gradient formula: $\frac{-5-7}{2-(-4)}$ **[M1]**

-2 **[A1]**

(b) Sets up the equation: $\frac{c-7}{0-(-4)} = -2$ **[M1]**

-1 **[A1]**

[Total Marks: 4]

5 (a) Uses difference in y over difference in x, e.g., $\frac{4-(-2)}{1-(-3)}$ **[M1]**

$\frac{3}{2}$ or 1.5 **[A1]**

(b) $\frac{-7-5}{4-(-2)}$ or $\frac{5-(-7)}{-2-4}$ **[M1]**

-2 **[A1]**

[Total Marks: 4]

6 (a) Correct vertical and horizontal change seen, e.g. $\frac{6}{4}$ or $\frac{3}{2}$ **[M1]**

1.5 (or $\frac{3}{2}$) **[A1]**

(b) Correct vertical and horizontal change seen, e.g. $\frac{-3}{6}$ **[M1]**

-0.5 (or $-\frac{1}{2}$) **[A1]**

(c) $\frac{-9-7}{5-(-3)}$ **[M1]**

-2 **[A1]**

[Total Marks: 6]

7 (a) (i) $\frac{3k-1-k}{5-2}$ **[M1]**

$\frac{2k-1}{3}$ **[A1]**

(ii) $\frac{2k-1}{3} = 4$ or $2k-1=12$ **[M1]**

6.5 or $\frac{13}{2}$ **[A1]**

[Total Marks: 4]

8 (a) $\frac{k-5}{4-(-2)} = -\frac{1}{2}$ **[M1]**

$k-5=-3$ **[M1]**

2 **[A1]**

(b) Gradient of $BC = \frac{-1-2}{10-4} = -\frac{1}{2}$ **[M1]**

Gradient of AB = gradient of BC and point B is common to both segments **[A1]**

(c) $\frac{(3p-1)-5}{p-(-2)} = 2$ **[M1]**

$3p-6=2(p+2)$ **[M1]**

10 **[A1]**

[Total Marks: 8]

9 (a) $\frac{(a+4)-3a}{5a-2a}$ or equivalent **[M1]**

$\frac{4-2a}{3a}$ **[A1]**

(b) States or implies gradient of PQ is $-\frac{1}{3}$ **[B1]**

$\frac{4-2a}{3a} = -\frac{1}{3}$ or $3(4-2a) = -3a$ **[M1]**

4 **[A1]**

(c) Gradient $AB = \frac{8-2}{3-(-1)}$ or $\frac{6}{4}$ or $\frac{3}{2}$ **[M1]**

$\frac{23-8}{c-3} = \frac{3}{2}$ or $\frac{23-2}{c-(-1)} = \frac{3}{2}$ **[M1]**

13 **[A1]**

[Total Marks: 8]

10 (a) $\frac{(a+6)-3a}{4-(-2)}$ **[M1]**

$\frac{6-2a}{6}$ **[M1]**

$\frac{3-a}{3}$ or $1-\frac{a}{3}$ **[A1]**

(b) $\frac{3-a}{3} = 3$ oe **[M1]**

$a = -6$ **[A1]**

(c) $P = (-2, -18)$ and $Q = (4, 0)$ **[M1]**

$\left(\frac{-2+4}{2}, \frac{-18+0}{2}\right)$ **[M1]**

$(1, -9)$ **[A1]**

[Total Marks: 8]

11 (a) $\frac{k-2}{4}$ **[B1]**

(b) Finds an expression for the gradient of BC (e.g., $\frac{k}{k-5}$) and equates to the gradient of AB: $\frac{k-2}{4} = \frac{k}{k-5}$ **[M1]**

Expands brackets correctly: $(k-2)(k-5) = 4k$

$k^2 - 7k + 10 = 4k$ [M1]

Rearranges to the given equation $k^2 - 11k + 10 = 0$ with no mathematical errors seen **[A1]**

(c) Factorises $(k-1)(k-10) = 0$ or uses the quadratic formula **[M1]**

1 and 10 **[A1]**

(d) Deduces that $k = 10$ must be used, as $k = 1$ would mean points A and C are the same, contradicting that they are distinct **[B1]**

Calculates the gradient of OB using $(5, 10)$: $\frac{10-0}{5-0} = 2$ **[B1]**

[Total Marks: 8]

12 (a) Calculates the gradient of AB as $\frac{9-3}{4-1} = 2$ and equates to gradient of BC or AC, e.g., $\frac{y-9}{8-4} = 2$ **[M1]**

$y = 17$ **[A1]**

(b) Sets up the gradient equation: $\dfrac{(p^2-p-3)-3}{p-1}=2$ **[M1]**

Simplifies to form a 3-term quadratic equation: $p^2-p-6=2(p-1)$

$\mathrm{p}^2-3p-4=0$ [M1]

Factorises $(p-4)(p+1)=0$ or uses the quadratic formula correctly **[M1]**

$p=4$ and $p=-1$ **[A1]**

(c) Sets up the gradient equation: $\dfrac{12-m}{(m+1)-3}=m$ **[M1]**

Multiplies by the denominator: $12-m=m(m-2)$ **[M1]**

Rearranges into a 3-term quadratic: $m^2-m-12=0$ **[M1]**

$m=4$ and $m=-3$ **[A1]**

[Total Marks: 10]

13 (a) $\dfrac{5-1}{4-(-2)}$ or equivalent **[M1]**

$\dfrac{2}{3}$ **[A1]**

(b) $\dfrac{-2-k}{8-3}=-1.4$ or equivalent **[M1]**

$-2-k=-7$ **[M1]**

5 **[A1]**

[Total Marks: 5]

14 (a) $\dfrac{-9-5}{4-(-3)}$ or $\dfrac{5-(-9)}{-3-4}$ **[M1]**

-2 **[A1]**

(b) $\dfrac{1-5}{k-(-3)}=-2$ oe **[M1]**

$-4=-2(k+3)$ or $-2k-6=-4$ **[M1]**

$k=-1$ **[A1]**

[Total Marks: 5]

15 (a) Forms a correct equation for the gradient: $\dfrac{5-2k}{2-(k-3)}=-3$ OR $\dfrac{2k-5}{k-3-2}=-3$ **[M1]**

Correctly simplifies the denominator and equates: $5-2k=-3(5-k)$ **[M1]**

$k=4$ **[A1]**

(b) Evaluates coordinates of P as $(1,8)$ and substitutes into midpoint formula:
$\left(\dfrac{1+2}{2},\dfrac{8+5}{2}\right)$ **[M1]**

$(1.5, 6.5)$ **[A1]**

[Total Marks: 5]

16 **(a)** Finds a correct algebraic expression for the gradient: $\frac{8-4}{k-2-3k}$ or $\frac{4}{-2k-2}$ **[M1]**

Equates to $-\frac{1}{2}$ and eliminates fractions: $8 = 2k + 2$ or $-8 = -2k - 2$ **[M1]**

3 **[A1]**

(b) $P(9, 4)$ and $Q(1, 8)$ **[B1]**

(c) Identifies the coordinates of R must be of the form $(0, y)$ **[B1]**

Sets up the gradient equation: $\frac{y-4}{0-9} = 1$ **[M1]**

$(0, -5)$ **[A1]**

[Total Marks: 7]

17 **(a)** Calculates gradient of AB: $\frac{8-(-4)}{3-(-1)} = \frac{12}{4} = 3$ **[B1]**

Sets up equation for gradient involving C, e.g., $\frac{23-8}{p-3} = 3$ or $\frac{23-(-4)}{p-(-1)} = 3$ **[M1]**

8 **[A1]**

(b) Forms expression for gradient AD: $\frac{2q-1-(-4)}{q-(-1)}$ **[M1]**

Equates to $-\frac{1}{3}$ and simplifies: $3(2q + 3) = -1(q + 1)$ **[M1]**

$-\frac{10}{7}$ or -1.43 (accept $-1.428...$) **[A1]**

[Total Marks: 6]

18 **(a)** Gradient $= \frac{(t^2 - 3t + 2) - 0}{t - 1}$ **[M1]**

Factorising numerator: $\frac{(t-1)(t-2)}{t-1}$ **[M1]**

Cancelling $(t - 1)$ correctly to obtain $t - 2$ **[A1]**

(b) $t - 2 = 2.8$ giving $t = 4.8$ **[M1]**

$y = 4.8^2 - 3(4.8) + 2$ **[M1]**

$(4.8, 10.64)$ **[A1]**

(c) Gradient of $QR = \frac{(t^2 - 3t + 2) - (-1.5)}{t - 3.2}$ **[M1]**

$\frac{t^2 - 3t + 3.5}{t - 3.2} = t - 2$ **[M1]**

$t^2 - 3t + 3.5 = t^2 - 5.2t + 6.4$ leading to $2.2t = 2.9$ **[M1]**

1.32 **[A1]**

[Total Marks: 10]